Limită punctuală: pentru orice $x \in [a, b]$, atunci:

$$\lim_{n \to \infty} |L_n f(x) - f(x)| = 0.$$

adică: pentru orice $x \in [a, b]$ şi orice $\varepsilon > 0$ există un $N(x, \varepsilon) \in \mathbb{N}$ astfel încât:

$$|L_n f(x) - f(x)| < \varepsilon$$

dacă $n > N(x, \varepsilon)$.

Limită uniformă:

$$\begin{aligned}
\lim_{n \to \infty} \max_{x \in [a,b]} |L_n f(x) - f(x)| &= 0 \\
\lim_{n \to \infty} \|L_n f - f\|_u &= 0 \\
L_n f &\rightrightarrows f
\end{aligned}$$

$$\|L_n f - f\|_u = \max_{x \in [a,b]} |L_n f(x) - f(x)|$$

adică pentru orice $\varepsilon > 0$ există un $N(\varepsilon) \in \mathbb{N}$ astfel încât:

$$|L_n f(x) - f(x)| < \varepsilon$$

dacă $n > N(\varepsilon)$, pentru orice $x \in [a, b]$.

Example 1 *şirul $(e_n)_{n \in N}$ converge punctual către funcţia $f : [0, 1] \to \mathbb{R}$ ($e_n(x) = x^n$):*

$$f(x) = \begin{cases} 0, & \text{dacă } x \in [0, 1) \\ 1, & \text{dacă } x = 1. \end{cases}$$

şi nu converge uniform către f.

Polinomul lui S.N. Bernstein (1912):

$$\begin{aligned}
B_n(f)(x) &= \sum_{k=0}^{n} \binom{n}{k} x^k (1 - x)^{n-k} f\left(\frac{k}{n}\right) \\
1 &= ((1 - x) + x)^n = \sum_{k=0}^{n} \binom{n}{k} x^k (1 - x)^{n-k}
\end{aligned}$$

obs.:

$$B_n(e_0)(x) = 1 = e_0(x)$$

Curbe si suprafeţe Bézier (60'). CAGD

Curbă Bézier pentru punctele $M_k(x_k, y_k)$, $k = \overline{0, n}$. Curba Bézier corespunzătoare punctelor M_k :

$$\begin{aligned}
Bezier(M_k, t) &= \sum_{k=0}^{n} \binom{n}{k} t^k (1 - t)^{n-k} (x_k, y_k). \\
x(t) &= \sum_{k=0}^{n} \binom{n}{k} t^k (1 - t)^{n-k} x_k \\
y(t) &= \sum_{k=0}^{n} \binom{n}{k} t^k (1 - t)^{n-k} y_k, t \in [0, 1].
\end{aligned}$$

Cele mai des folosite sunt c. Bezier pentru $n = 3$ (4 puncte).

1. Fie $\Delta_n : 0 = x_0 < x_1 < \ldots < x_n = 1$ un şir de diviziuni a intervalului $[0, 1]$ şi $s_{\Delta_n}(f)$, $n \in \mathbb{N}^*$ splinul de interpolare de ordinul întâi. Să se arate că dacă $\|\Delta_n\| \to 0$ atunci

$$\lim_{n\to\infty} \|s_{\Delta_n}(f) - f\|_\infty = 0,$$

pentru orice funcţie continuă pe $[0, 1]$.

Ideea: PBK adică arăt că

$$\lim_{\substack{n\to\infty \\ \|\Delta_n\|\to 0}} \|s_{\Delta_n}(e_i) - e_i\| = 0, i = 0, 1, 2.$$

$$\begin{aligned} s_{\Delta_n}(e_0)(x) &= e_0(x) = 1 \\ s_{\Delta_n}(e_1)(x) &= e_1(x) = x \end{aligned}$$

$$s_{\Delta_n}(e_2)(x) = ?$$

pentru $x \in [x_i, x_{i+1}]$:

$$\begin{aligned} s_{\Delta_n}(e_2)(x) &= x_i^2 + \frac{x_{i+1}^2 - x_i^2}{x_{i+1} - x_i}(x - x_i) \\ \left|s_{\Delta_n}(e_2)(x) - x^2\right| &= \left|x_i^2 + \frac{x_{i+1}^2 - x_i^2}{x_{i+1} - x_i}(x - x_i) - x^2\right| \overset{|x_{i+1}-x_i|\to 0}{\longrightarrow} 0 \end{aligned}$$

$$\begin{aligned} \left|x_i^2 + \frac{x_{i+1}^2 - x_i^2}{x_{i+1} - x_i}(x - x_i) - x^2\right| &= |-(x - x_{i+1})(x - x_i)| \le \\ &\le \left|\frac{(x_{i+1} - x_i)^2}{4}\right| \le \|\Delta_n\|^2 / 4 \end{aligned}$$

$$(x - x_{i+1})\big|_{x=\frac{x_i + x_{i+1}}{2}} = \frac{x_i - x_{i+1}}{2}$$

Deci:

$$\|s_{\Delta_n}(e_2) - e_2\| \le \|\Delta_n\|^2 / 4$$

de unde:

$$s_{\Delta_n}(e_2) \overset{\|\Delta_n\|\to 0}{\rightrightarrows} e_2$$

Problemă: $(D_n)_{n\in N}$ este un şir de aproximare uniformă:

$$D_n(f)(x) = (n+1)\sum_{k=0}^{n} p_{n,k}(x)\int_0^1 p_{n,k}(t)f(t)dt.$$

Ideea: Teorema PBK.

$$D_n\left(f\right)\left(x\right) = \left(n+1\right)\sum_{k=0}^{n}\binom{n}{k}x^k\left(1-x\right)^{n-k}\int_0^1\binom{n}{k}t^k\left(1-t\right)^{n-k}f\left(t\right)dt$$

arătăm că

$$\lim_{n\to\infty}\max_{x\in[0,1]}\left|D_n\left(e_i\right)\left(x\right)-e_i\left(x\right)\right| = 0, i = 0,1,2,$$

$$\begin{aligned}
D_n\left(e_0\right)\left(x\right) &= \left(n+1\right)\sum_{k=0}^{n}\binom{n}{k}x^k\left(1-x\right)^{n-k}\int_0^1\binom{n}{k}t^k\left(1-t\right)^{n-k}dt \overset{SOC}{=} \\
&= \left(n+1\right)\sum_{k=0}^{n}\binom{n}{k}x^k\left(1-x\right)^{n-k}\binom{n}{k}B\left(k+1,n-k+1\right)
\end{aligned}$$

unde $B\left(\alpha,\beta\right)$ este funcţia Beta a lui Euler:

$$B\left(\alpha,\beta\right) = \int_0^1 t^{\alpha-1}\left(1-t\right)^{\beta-1}dt$$

Se ştie că

$$\begin{aligned}
B\left(\alpha,\beta\right) &= \frac{\Gamma\left(\alpha\right)\Gamma\left(\beta\right)}{\Gamma\left(\alpha+\beta\right)}, \\
\Gamma\left(p+1\right) &= p!, p\in\mathbb{N}
\end{aligned}$$

Atunci:

$$\begin{aligned}
D_n\left(e_0\right)\left(x\right) &= \left(n+1\right)\sum_{k=0}^{n}\binom{n}{k}x^k\left(1-x\right)^{n-k}\binom{n}{k}\frac{\Gamma\left(k+1\right)\Gamma\left(n-k+1\right)}{\Gamma\left(n-k+k+2\right)} = \\
&= \left(n+1\right)\sum_{k=0}^{n}\binom{n}{k}x^k\left(1-x\right)^{n-k}\frac{n!}{k!\left(n-k\right)!}\frac{k!\left(n-k\right)!}{\left(n+1\right)!} = \\
&= \left(n+1\right)\sum_{k=0}^{n}\binom{n}{k}x^k\left(1-x\right)^{n-k}\frac{1}{n+1} = \sum_{k=0}^{n}\binom{n}{k}x^k\left(1-x\right)^{n-k} \\
&= \left(\left(1-x\right)+x\right)^n = e_0\left(x\right)
\end{aligned}$$

Analog:

$$
\begin{aligned}
D_n\left(e_1\right)(x) &= (n+1)\sum_{k=0}^{n}\binom{n}{k}x^k\left(1-x\right)^{n-k}\int_0^1\binom{n}{k}t^k\left(1-t\right)^{n-k}tdt = \\
&= (n+1)\sum_{k=0}^{n}\binom{n}{k}x^k\left(1-x\right)^{n-k}\binom{n}{k}B\left(k+2,n-k+1\right) = \\
&= (n+1)\sum_{k=0}^{n}\binom{n}{k}x^k\left(1-x\right)^{n-k}\frac{n!}{k!\left(n-k\right)!}\frac{\left(k+1\right)!\left(n-k\right)!}{\left(n-k+k+2+1-1\right)!} = \\
&= (n+1)\sum_{k=0}^{n}\binom{n}{k}x^k\left(1-x\right)^{n-k}\frac{k+1}{\left(n+1\right)\left(n+2\right)} = \\
&= \frac{1}{n+2}\sum_{k=0}^{n}\binom{n}{k}\left(k+1\right)x^k\left(1-x\right)^{n-k} = \\
&= \frac{1}{n+2}\left(\sum_{k=0}^{n}\binom{n}{k}kx^k\left(1-x\right)^{n-k}+\sum_{k=0}^{n}\binom{n}{k}x^k\left(1-x\right)^{n-k}\right) = \\
&= \frac{1}{n+2}\left(\sum_{k=1}^{n}\frac{n!}{\left(k-1\right)!\left(n-k\right)!}x^k\left(1-x\right)^{n-k}+1\right) = \\
&= \frac{1}{n+2}x\sum_{k=1}^{n}\frac{n\left(\left(n-1\right)!\right)}{\left(k-1\right)!\left(n-1-\left(k-1\right)\right)!}x^{k-1}\left(1-x\right)^{n-1-(k-1)}+\frac{1}{n+2} = \\
&= \frac{nx}{n+2}\left(\left(1-x\right)+x\right)^{n-1}+\frac{1}{n+2} = \frac{nx}{n+2}+\frac{1}{n+2}\overset{n\to\infty}{\to}x, \text{ uniform:}
\end{aligned}
$$

$$
\begin{aligned}
\left|D_n\left(e_1\right)(x)-x\right| &= \left|\frac{nx}{n+2}+\frac{1}{n+2}-x\right| = \\
&= \left|-\frac{2x}{n+2}+\frac{1}{n+2}\right| \leq \frac{3}{n+2}\to 0
\end{aligned}
$$

Analog

$$\begin{aligned}
D_n(e_2)(x) &= (n+1)\sum_{k=0}^{n}\binom{n}{k}x^k(1-x)^{n-k}\int_0^1\binom{n}{k}t^k(1-t)^{n-k}t^2dt = \\
&= (n+1)\sum_{k=0}^{n}\binom{n}{k}x^k(1-x)^{n-k}\binom{n}{k}B(k+3,n-k+1) = \\
&= (n+1)\sum_{k=0}^{n}\binom{n}{k}x^k(1-x)^{n-k}\frac{n!}{k!(n-k)!}\frac{(k+2)!(n-k)!}{(n-k+k+3)!} = \\
&= (n+1)\sum_{k=0}^{n}\binom{n}{k}x^k(1-x)^{n-k}\frac{(k+2)(k+1)}{(n+1)(n+2)(n+3)} = \\
&= \frac{1}{(n+2)(n+3)}\sum_{k=0}^{n}\binom{n}{k}\left(k^2+3k+2\right)x^k(1-x)^{n-k}\overset{SOC}{=} \\
&= \frac{1}{(n+2)(n+3)}\left(n^2\sum_{k=0}^{n}\binom{n}{k}\frac{k^2}{n^2}x^k(1-x)^{n-k}+3n\sum_{k=0}^{n}\binom{n}{k}\frac{k}{n}x^k(1-x)^{n-k}+2\cdot 1^n\right) = \\
&= \frac{n^2}{(n+2)(n+3)}B_n(e_2)(x)+\frac{3n}{(n+2)(n+3)}B_n(e_1)(x)+\frac{2}{(n+2)(n+3)}\rightrightarrows e_2\blacksquare
\end{aligned}$$

2. Să se calculeze $B_n(e_3)(x)$.

$$\begin{aligned}
B_n(e_3)(x) &= \sum_{k=0}^{n}\binom{n}{k}x^k(1-x)^{n-k}\frac{k^3}{n^3} = \\
&= \frac{1}{n^3}\sum_{k=0}^{n}\binom{n}{k}k^3x^k(1-x)^{n-k}
\end{aligned}$$

ideea: se foloseşte funcţia ajutătoare

$$\begin{aligned}
g(x,t) &= \left(1-x+xe^t\right)^n = \sum_{k=0}^{n}\binom{n}{k}e^{kt}x^k(1-x)^{n-k} \\
\frac{\partial^3 g}{\partial t^3}(x,t) &= \sum_{k=0}^{n}\binom{n}{k}k^3e^{kt}x^k(1-x)^{n-k} \\
\frac{\partial^3 g}{\partial t^3}(x,0) &= \sum_{k=0}^{n}\binom{n}{k}k^3x^k(1-x)^{n-k}
\end{aligned}$$

dar:

$$\begin{aligned}
\frac{\partial g}{\partial t}(x,t) &= xe^t n\left(1-x+xe^t\right)^{n-1} \\
\frac{\partial^2 g}{\partial t^2}(x,t) &= xe^t n\left(1-x+xe^t\right)^{n-1} + n(n-1)x^2e^{2t}\left(1-x+xe^t\right)^{n-2} \\
\frac{\partial^3 g}{\partial t^3}(x,t) &= xe^t n\left(1-x+xe^t\right)^{n-1} + 3n(n-1)x^2e^{2t}\left(1-x+xe^t\right)^{n-2} + \\
&\quad + n(n-1)(n-2)x^3e^{3t}\left(1-x+xe^t\right)^{n-3}
\end{aligned}$$

deci

$$\frac{\partial^3 g}{\partial t^3}(x,0) = xn + 3n(n-1)x^2 + n(n-1)(n-2)x^3$$

Deci

$$B_n(e_3)(x) = \frac{xn + 3n(n-1)x^2 + n(n-1)(n-2)x^3}{n^3}$$